

DIGITAL FORENSICS AND INCIDENT RESPONSE SIMULATION

Evidence Acquisition, Analysis & Response Playbook Development Report

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EXECUTIVE SUMMARY

This report documents a complete digital forensics and incident response simulation conducted on a deliberately compromised Windows 10 virtual machine within the isolated cybersecurity home laboratory. The exercise replicated a realistic post-exploitation scenario from the authorised penetration test (Project #2), including webshell persistence, credential access, and simulated data exfiltration.

The full forensic lifecycle was executed: evidence acquisition (disk E01 image and raw memory dump), analysis using industry-standard tools (Autopsy for disk forensics and Volatility for memory forensics), timeline reconstruction, professional reporting, and development of containment/eradication/recovery playbooks. All activities were performed ethically with full chain-of-custody documentation and snapshot rollback capability.

Key outcomes include a forensically sound evidence acquisition process, detailed attack timeline reconstruction, identification of artefacts missed by disk-only analysis (memory injection, network connections), and actionable response playbooks. This work provides direct evidence of investigative and incident response capability aligned with the responsibilities of a Navy Information Warfare Officer.

1. INTRODUCTION AND OBJECTIVES

The RAAF Cyber Warfare Officer role requires not only the ability to prevent and detect attacks but also to investigate incidents, preserve evidence, reconstruct attacker actions, and lead effective response. This simulation builds directly on the lab environment (Project #1), authorised pentest (Project #2), and SIEM deployment (Project #3) by adding the critical post-incident investigation and response layer.

Objectives:

- Create a realistic compromised state on a Windows 10 VM replicating post-exploitation from Project #2.
- Perform forensically sound evidence acquisition (disk image + memory dump) with hash verification.
- Conduct full analysis using Autopsy (disk) and Volatility (memory) to reconstruct the attack timeline.
- Produce a professional forensic report suitable for ADF/government consumption.
- Develop practical containment, eradication, and recovery playbooks.

2. DETAILED IMPLEMENTATION AND PRACTICAL STEPS

2.1 Incident Simulation Setup

A clean Windows 10 VM was reverted to a known-good snapshot. The following artefacts were deliberately introduced to simulate post-exploitation from the webshell attack in Project #2:

- Uploaded a PHP webshell (webshell.php) to C:\inetpub\wwwroot\
- Created persistence via Run registry key (HKCU\Software\Microsoft\Windows\CurrentVersion\Run)
- Executed suspicious PowerShell commands and left event log artefacts
- Created and "exfiltrated" a fake sensitive document (C:\Users\Public\sensitive_data.txt)
- Left a hidden malicious process running (simulated via cmd.exe with suspicious parent)

2.2 Evidence Acquisition

Disk Image Acquisition: The VM was cleanly shut down. Using FTK Imager on the host:

1. Created a new case and added the virtual hard disk (.vmdk or .vhdx).
2. Selected "Create Image" → Physical → Calculate MD5 + SHA1 hashes.
3. Output format: E01 (Expert Witness) with compression and verification.

Command-line alternative used for verification:

```
ftkimager.exe PhysicalDrive2 E:\Evidence\Win10_Incident_2026-07-06 --e01 --verify
```

Memory Acquisition: Before shutdown, a raw memory dump was captured using WinPMEM or DumpIt.exe:

```
DumpIt.exe /OUTPUT E:\Evidence\Win10_Memory_2026-07-06.raw
```

Both images were hashed and logged with timestamps and chain-of-custody notes.

2.3 Disk Forensics with Autopsy

A new case was created in Autopsy 4.x:

- 1. Case name: Lab_Incident_2026-07-06
- 2. Added the E01 image as data source.
- 3. Enabled all ingest modules: File Type Identification, Extension Mismatch, Embedded Metadata, Encryption Detection, Timeline, Web Artifacts, Recent Activity, Hash Lookup (NSRL + custom bad hashes).

Key findings in Autopsy:

- Webshell.php located in C:\inetpub\wwwroot\ with creation timestamp matching attack window.
- Registry Run key pointing to suspicious executable (persistence).
- PowerShell execution artefacts and prefetch files showing suspicious activity.
- Timeline view clearly showed the sequence: webshell upload → registry modification → file access → exfiltration.

2.4 Memory Forensics with Volatility

The raw memory image was analysed using Volatility 3 (or 2.6.1 for compatibility):

```
vol.py -f Win10_Memory_2026-07-06.raw imageinfo # Identify profile (Win10x64_19041)
vol.py -f Win10_Memory_2026-07-06.raw --profile=Win10x64_19041 pslist
vol.py -f Win10_Memory_2026-07-06.raw --profile=Win10x64_19041 malfind # Detected injected code
vol.py -f Win10_Memory_2026-07-06.raw --profile=Win10x64_19041 netscan # Showed suspicious outbound connection
vol.py -f Win10_Memory_2026-07-06.raw --profile=Win10x64_19041 filescan | grep sensitive
```

Memory analysis revealed artefacts invisible on disk: code injection into legitimate processes, network connections that had already closed, and command history not written to disk. These findings were critical to completing the attack timeline.

2.5 Timeline Reconstruction and Reporting

A consolidated timeline was built combining Autopsy timeline output, Volatility process/network data, and Windows event logs. The attack sequence was reconstructed minute-by-minute, showing initial foothold, persistence installation, credential access, and data staging/exfiltration. A professional forensic report was authored with executive summary, technical findings, artefact tables, and timeline visualisation suitable for command-level review.

Time (approx)	Event / Artefact	Source	MITRE
T+0 min	Webshell upload (webshell.php created)	Autopsy MFT / Timeline	T1190
T+2 min	Registry Run key persistence added	Autopsy Registry	T1547.001
T+5 min	PowerShell execution + suspicious process	Event Logs + Volatility pslist	T1059.001
T+8 min	sensitive_data.txt accessed & staged	Autopsy + Volatility filescan	T1005
T+12 min	Outbound connection to attacker IP	Volatility netscan	T1071

Table 1: Reconstructed Attack Timeline with Sources and MITRE ATT&CK; Mapping

2.6 Response Playbook Development

Three practical playbooks were developed and documented:

- **Containment Playbook:** Isolate host (network quarantine via pfSense), preserve volatile data (memory dump), block attacker IP at firewall.
- **Eradication Playbook:** Remove webshell and persistence mechanisms, reset compromised credentials, patch exploited vulnerabilities, re-image or restore from known-good snapshot.
- **Recovery Playbook:** Restore from clean snapshot, validate with Autopsy/Volatility baseline comparison, enhance monitoring rules in SIEM, conduct lessons-learned review.

3. LESSONS LEARNED AND ADF RELEVANCE

This simulation provided several operationally relevant insights:

- **Memory forensics is essential:** Several critical artefacts (injected code, closed network connections, command history) existed only in memory. Disk-only forensics would have produced an incomplete picture.
- **Timeline reconstruction is powerful for leadership:** A clear, minute-by-minute attack timeline is far more useful to commanders than a list of artefacts. This skill is directly transferable to operational incident reporting.
- **Playbooks must be actionable and tested:** Generic checklists are insufficient. Playbooks need specific commands, tool locations, and decision points tailored to the environment.

- **Chain of custody and documentation are non-negotiable:** Even in a lab exercise, maintaining proper documentation and hash verification built habits essential for real incidents that may lead to legal or disciplinary action.

These capabilities — evidence acquisition, memory and disk forensics, timeline analysis, and response playbook development — map directly to the investigative and incident response responsibilities of a Cyber Warfare Officer protecting ADF networks and information systems.

4. CONCLUSION

The digital forensics and incident response simulation was successfully completed. Evidence was acquired forensically soundly, analysed using industry-standard tools, and used to reconstruct a realistic attack timeline. Professional reporting and practical response playbooks were produced. This work provides strong, hands-on evidence of investigative and response capability directly aligned with the RAAF Cyber Warfare Officer role. Future integration with the SIEM (Project #3) will enable automated alert-to-forensics workflows.

References: Autopsy Documentation, Volatility Framework Documentation, SANS DFIR resources, ASD ISM Incident Management sections, Microsoft Windows Forensics reference.